GenAI 10-Layers
FinQuery → Enterprise PoC

Technical Robustness & Organizational Readiness Blueprint

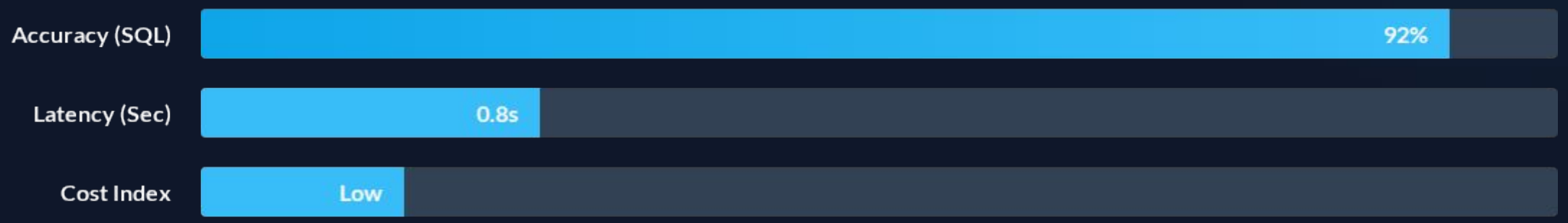
AUDIENCE: EXECUTIVES + DELIVERY

01. INFRASTRUCTURE FOUNDATION

- **Leverage** Nginix port :80 / :443
- SQLite with WAL mode (auto-enabled by SQLAlchemy)
- **Enable** auto-scaling to handle peak analyst loads.
- **Deploy** within VPC for secure internal access.
- **Automate** infrastructure with GitHub Actions.

02. FOUNDATION MODEL CHOICE

Llama.3.1 8B selected for superior speed-to-cost ratio in Text-to-SQL tasks.



- ✓ **Llama.3.1 8B** - 128K context length

03. SECURITY & GUARDRAILS

Auth Based Retrieval

Every generation is controlled by the User session and auth secret. if the user auth fails the LLM Will not generate an answer.

Input Guardrails

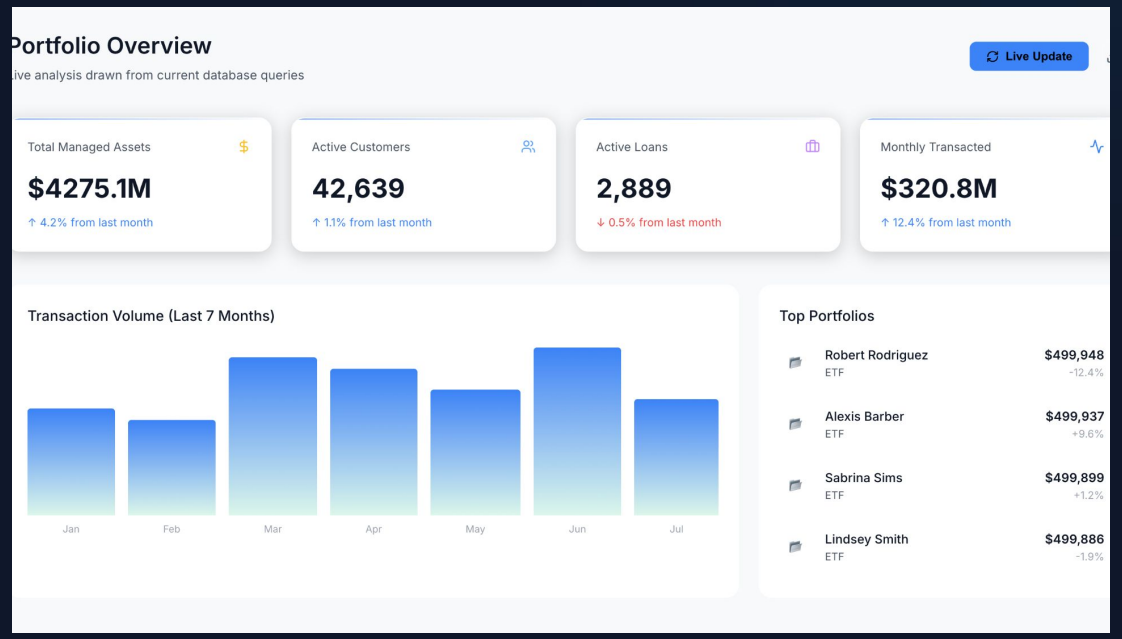
Sanitize natural language queries to prevent prompt injection attacks.

IAM Roles

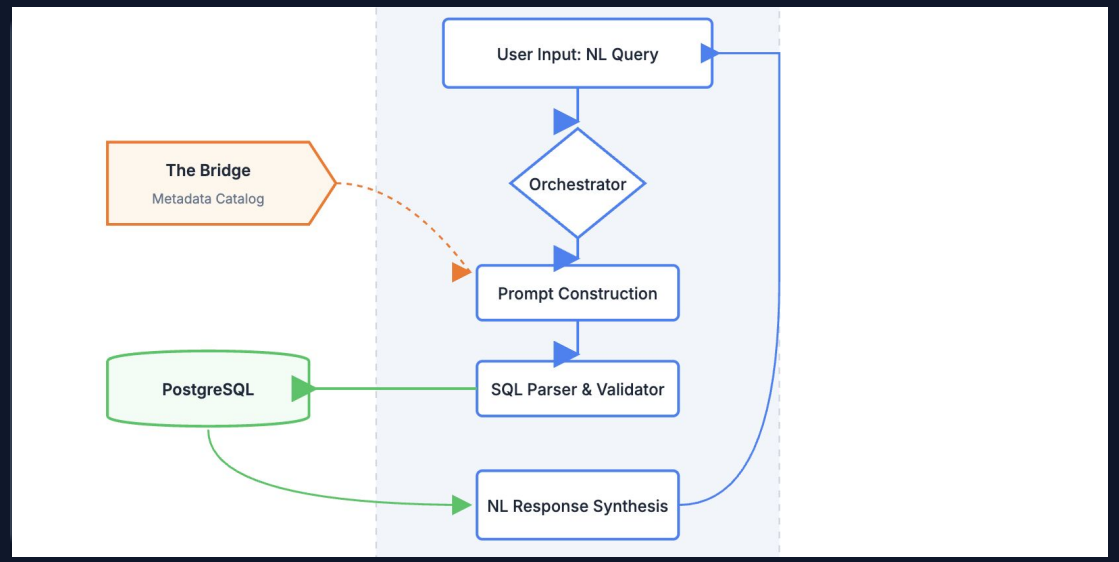
Isolate data access using least-privilege service accounts.

04. DATA: THE "METADATA BRIDGE"

- **Source** financial data from SQL tables.
- **Catalog** metadata (schema, types, relationships).
- **Embed** schema info in Vector DB for context retrieval.
- **Schedule** pipelines for daily data freshness.
- **Clean** noise to ensure SQL join accuracy.



05. APP & ORCHESTRATION FLOW



- **Interface**: React Chatbot Interface.
- **Middleware**: Django / LangChain Agent Orchestration
- **Cache**: Redis for common SQL results.
- **Observe**: Trace logs for every user query.
- **Retry**: Automated error correction on SQL fail.

05. APP & ORCHESTRATION FLOW

High-Level Architecture Components

Frontend Client

Streamlit web interface with secure auth. Captures NL queries and renders tables/charts from JSON results.

API Middleware

Django layer on Cloud Run. Handles rate limiting, user session context, and initial PII scrubbing via Cloud DLP.

Metadata Bridge

Vector store (Chroma/Qdrant) containing schema documentation, allowing the LLM to 'know' table relationships dynamically.

Orchestration Hub

LangChain execution engine. Manages few-shot prompt templates and handles self-healing retry logic for malformed SQL.

LLM Engine

Gemini 1.5 Flash via Vertex AI. High-speed, low-cost inference for SQL generation and final insight synthesis.

Secure Data Layer

Enterprise PostgreSQL instance. Accessed via a read-only service account with restricted schema scope.

06. SPONSORSHIP & STRATEGY

Exec Sponsor

Align with CFO's "Digital Finance" initiative for 2025.

Decision Rights

Define Finance as Lead Approver; IT as Quality Assurance.

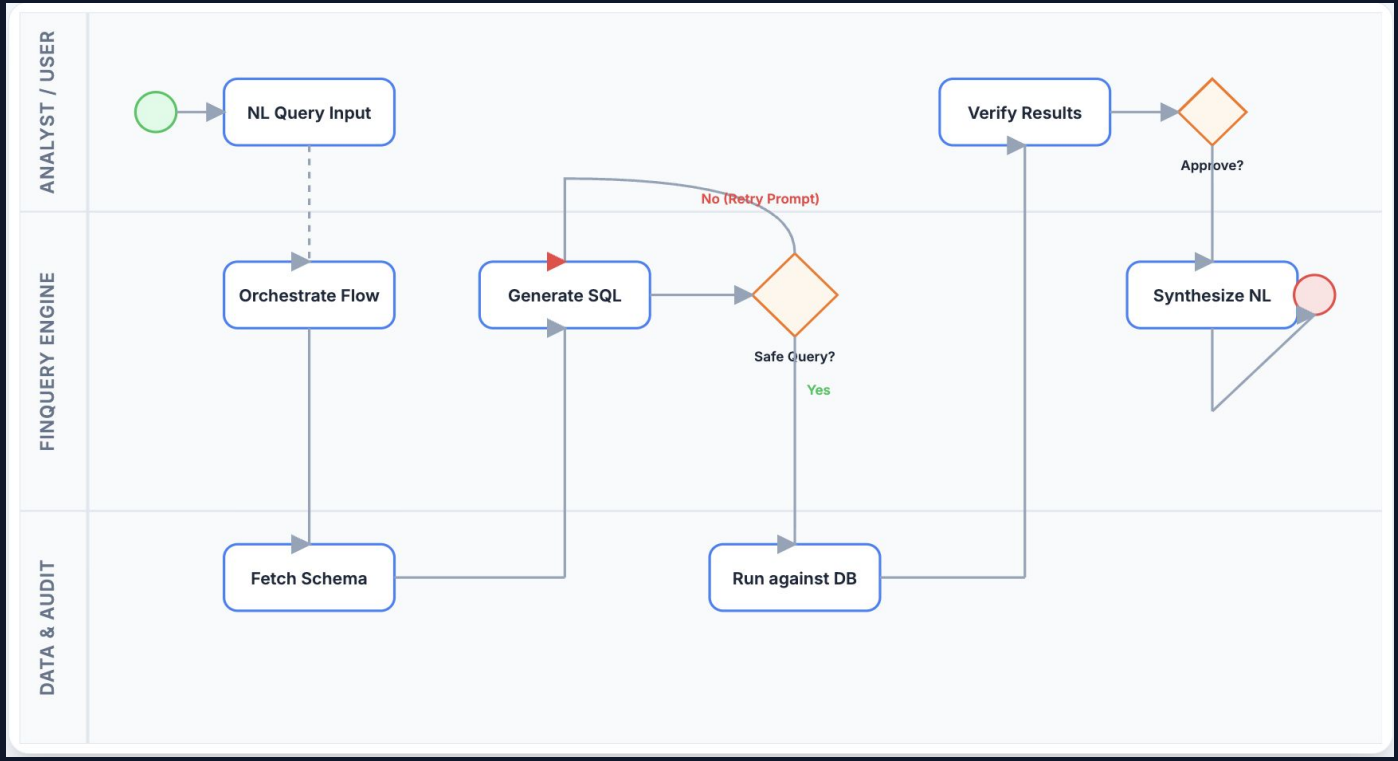
Funding

Allocate PoC budget via Innovation fund (Capex).

USER STORY	BUSINESS PROCESS	POC STAGE
"As an Account Manager, I want to query client revenue trends instantly so I can prepare for meetings without IT support."	Self-Service Analytics	Discovery & Discovery
"As a Financial Controller, I want to verify the SQL generated by the AI so I can ensure the numbers reported are compliant."	Quality Assurance (QA)	Technical Validation
"As a Data Engineer, I want to update the Metadata Bridge easily so the LLM remains aware of new schema changes."	Governance & Maintenance	Pipeline Integration
"As an Executive, I want to see a summary dashboard of time saved across teams so I can justify the scaling of this PoC."	Value Realization	Evaluation & Scaling

07. PROCESS & GOVERNANCE

Stage	Approval Gate	IT/Biz Task
Draft	Prompt Review	Validate SQL logic vs Finance logic
Testing	Hallucination Check	Audit AI outputs against "Golden Dataset"
Staging	Performance Gate	Check latency & error rate KPIs
Live	Periodic Audit	Review model registry every 30 days



08. ONBOARDING & TRAINING

- **Host** Prompt-Engineering labs for Analysts.
- **Conduct** Responsible AI awareness training.
- **Appoint** "AI Champions" in each business unit.
- **Release** FAQ & prompt-best-practices guide.
- **Establish** feedback channel for live tuning.

09. UX, KPIS & PERFORMANCE

85%
Time Saved per Query

<5%
Error Rate (SQL)

20-30
Users on POC

Next Steps Roadmap (M0 - M2)

M0: Architecture Setup & Security Approval | **M1:** Data Integration & Beta Testing | **M2:** Production Pilot & KPI Review

IMAGE SOURCES

https://d3nqmfhxc38njt.cloudfront.net/wp-content/uploads/2022/11/Architecture-of-Cloud-computing-930x620.png

Source: krazytech.com

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